Unit Testing and Integration Testing

Software testing is used to find errors and ensure the software works correctly. Two important levels of testing are:

- Unit Testing
- Integration Testing

These are done in early stages of development to improve software quality.

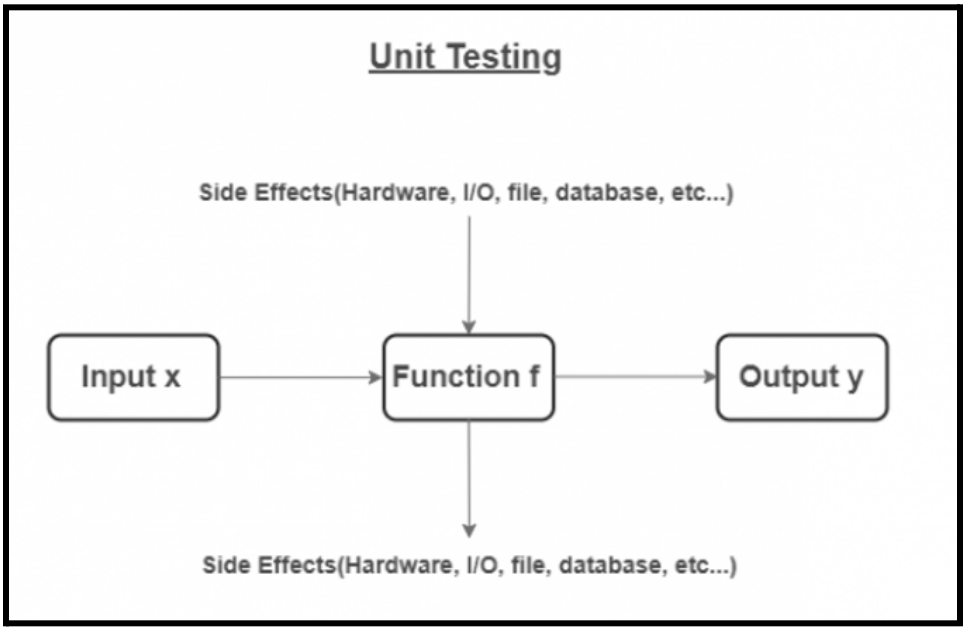
---

1. Unit Testing

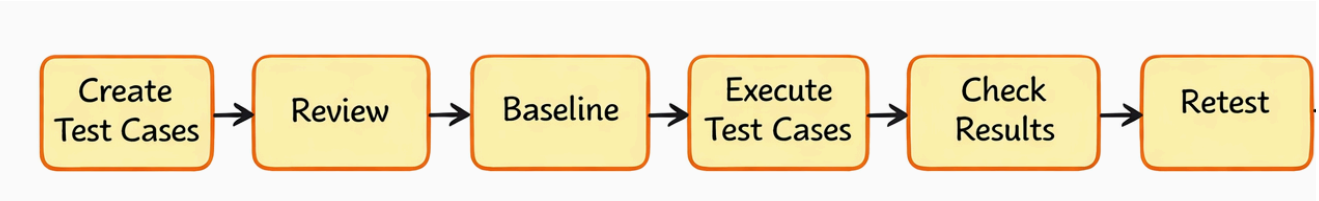
Unit testing is a type of testing where individual modules or components of software are tested separately. It focuses on small parts of the program.

Objectives of Unit Testing

- ❖ To test each function or method
- ❖ To verify correctness of code
- ❖ To detect bugs early
- ❖ To improve code quality
- ❖ To help developers understand the code



Workflow of Unit Testing



Types of Unit Testing Techniques

1. Black Box Testing

- Also called **behavioral testing**
- Focuses on input and output
- Test cases are derived from requirements
- Does not consider internal structure

Errors Found:

- Missing functions
- Interface errors
- Data structure errors
- Performance errors
- Initialization

---

2. White Box Testing

- Also called **structural testing**
- Tests internal structure of program
- Requires knowledge of code

Types:

a) Condition Testing

- Tests logical conditions in program
- Checks true/false conditions

b) Loop Testing

- Tests loops in program
- Types of loops:
  - Simple loop
  - Nested loop
  - Concatenated loop
  - Unstructured loop

Unit Testing Tools

- JUnit
- NUnit
- TestNG
- Emma
- PHPUnit

Advantages of Unit Testing

1) Helps developers learn code behavior
2) Ensures module works properly
3) Allows testing before full system completion
4) Early detection of errors
5) Improves code quality

Disadvantages of Unit Testing

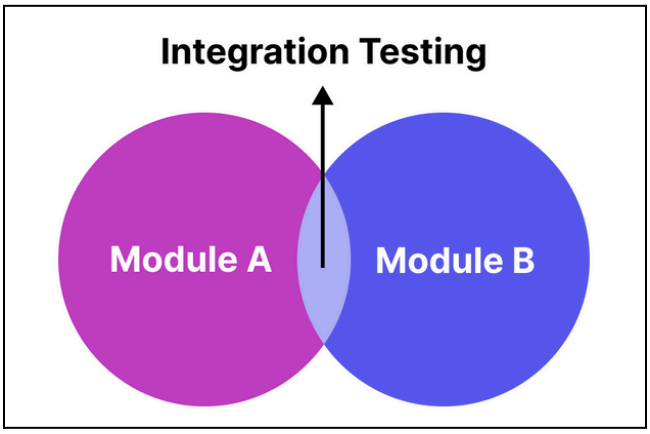
1. 1Time consuming
2. Cannot detect all errors
3. Errors may appear during integration

---

2. Integration Testing

1. Definition

Integration testing is the process of testing combined modules to verify their interaction. It focuses specifically on the interfaces between integrated units. This level of testing is designed to find errors in the way components work together.

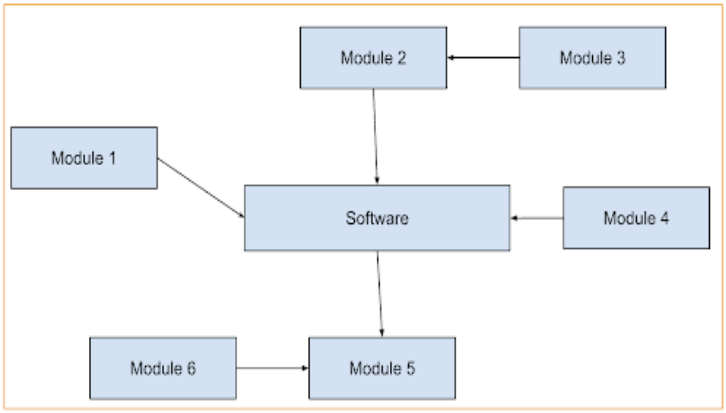


Purpose of Integration Testing

- To expose faults in interaction between modules
- To verify combined functionality
- To ensure proper data flow

Types of Integration Testing

1. Big Bang Integration Testing



- All modules are integrated at once
- Entire system tested as a **whole**

Advantages:

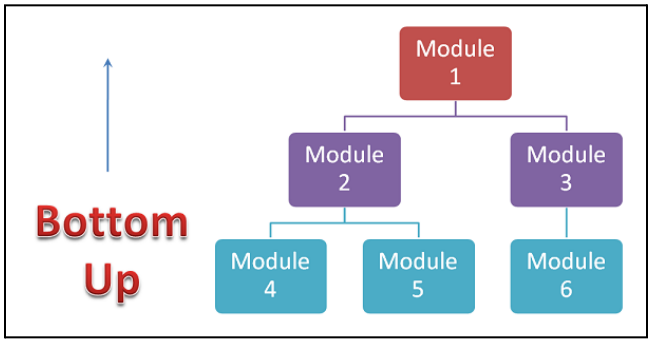
- Simple approach
- Suitable for small systems
- No planning require

Disadvantages:

- Difficult to locate errors
- Debugging is hard
- Errors may belong to any module

---

2. Bottom-Up Integration Testing



- Testing starts from **lower level modules**
- Modules are integrated step by step
- Uses **drivers**

Advantages:

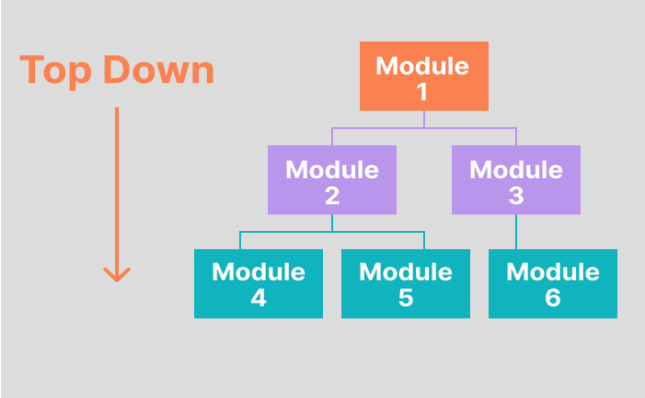
- No stubs required
- Easy to detect errors
- Suitable for bottom-up design

Disadvantages:

- Drivers must be developed
- High-level modules tested late

3. Top-Down Integration Testing

- Testing starts from **top modules**
- Uses **stubs**
- Lower modules added step by step



Advantages:

- Important modules tested early
- No drivers required
- Easy to find interface errors

Disadvantages:

- Requires many stubs
- Low-level modules tested later
- Difficult to design test cases

Regression Testing

Regression testing is the systematic process of verifying modified code and existing unchanged components to ensure that recent changes have not introduced new errors or regressions. This procedure is initiated when software modifications occur, including:

- Adding new features
- Fixing bugs
- Optimization

---

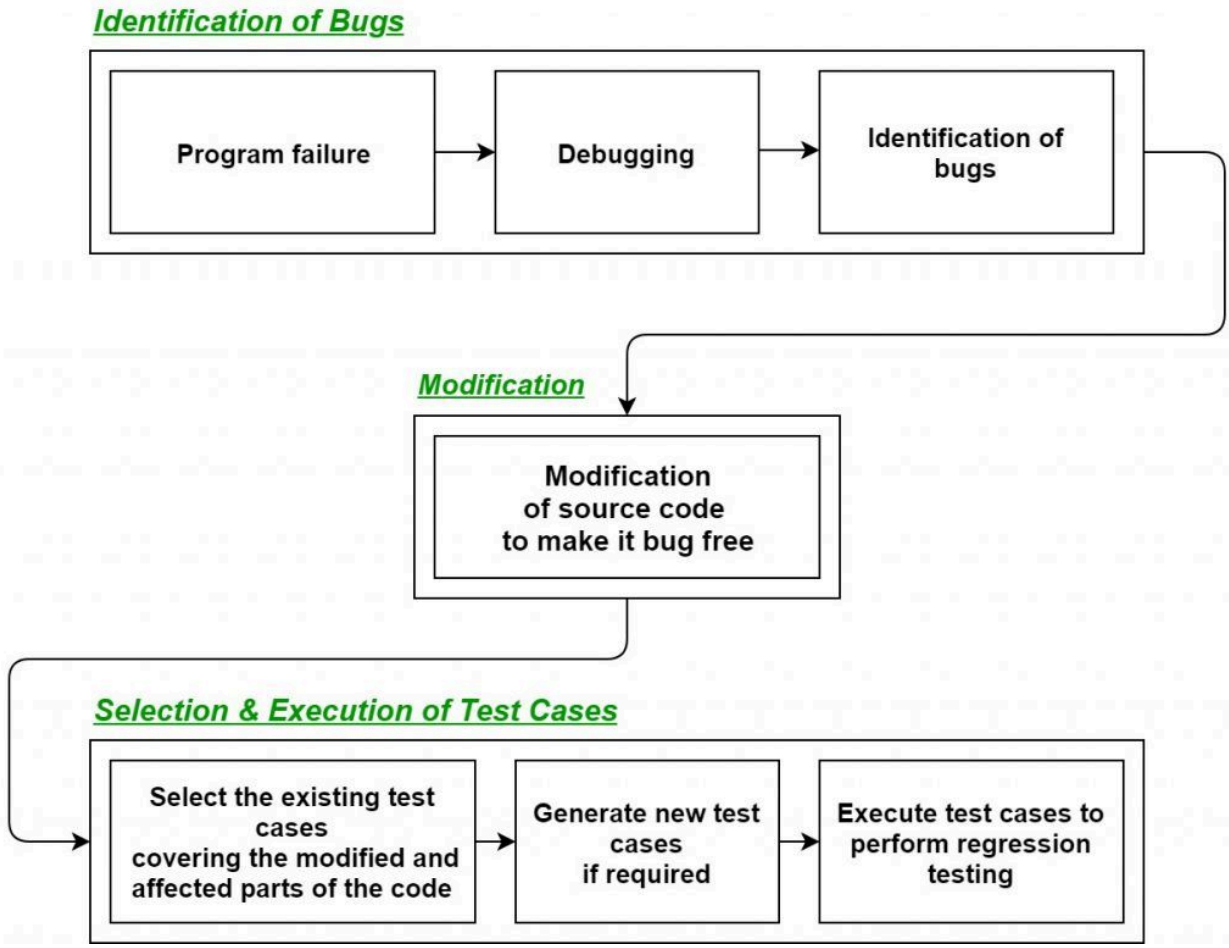
2. Purpose of Regression Testing

- To ensure modifications do not negatively impact existing functionality
- To verify that legacy test cases continue to execute correctly
- To identify new defects resulting from modifications

3. Process of Regression Testing

1) Modify the source code
2) Execute previously established test cases
3) Analyze results
4) Identify bugs
5) Perform debugging on the program

Identification of Targets, Select & Execute Test Cases



Prioritization of Test Cases

Based on Technical Requirements:

1) Essential test cases
2) Important test cases
3) Execute resource planning
4) Less important test cases
5) Redundant test cases

Regression Testing Tools

- Selenium
- Watir
- QTP
- RFT
- WinRunner
- SilkTest

Advantages of Regression Testing

1) Ensures that new modifications do not disrupt established features
2) Aids in maintaining overall software excellence
3) Detects structural defects post-modification

Disadvantages of Regression Testing

1) High consumption of time and resources
2) Necessity for automation tools in large systems
3) Lack of efficiency regarding minor adjustments